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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (MMD) (CBCS) Degree Examinations

May – June 2025

22MMD231 – COMPOSITE MATERIALS TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Derive the rule of mixtures for longitudinal modules of a unidirectional lamina using the strength of the material approach. State the assumptions to be made in the approach.	10	L*2	1	1
	b)	Explain with a schematics sketch, the manufacturing of glass fiber.	10	L2	2	2
2.	a)	Explain composite materials. How do you classify them?	06	L2	1	2
	b)	Explain the lamina and laminate.	04	L2	1	
	c)	Derive the major Poisson's ratio equation used in the composite materials.	10	L2	2	2
3.	a)	Discuss the role of fiber and matrix material in composites.	10	L2	1	1
	b)	A Glass/Epoxy lamina consists of a 70% fiber volume fraction. Use properties of glass and epoxy, respectively to determine the i) density of lamina ii) mass fractions of the glass and epoxy iii) volume of composite lamina if the mass of the lamina is 4 kg. iv) volume and mass of glass and epoxy in part (c).	10	L3	2	1
		Unit – II				
4.	a)	Explain the filament winding process with a neat sketch. List its merits and demerits.	10	L2	3	1
	b)	What are the NDT methods used in the testing of composite materials? Explain any one method.	10	L2	4	2
5.	a)	Analyze the maximum value of $S > 0$ if a stress $\sigma_x = 2S$, $\sigma_y = -3S$, and $\tau_{xy} = 4S$ are applied to a 60° lamina of graphite/epoxy. Use maximum stress theory and strain theory. Use the properties of a unidirectional graphite/epoxy lamina ($E_1 = 181 \text{ GPa}$, $E_2 = 10.3 \text{ GPa}$, $\nu_{12} = 0.28$, $G_{12} = 7.17 \text{ GPa}$, $(s_2^C)_{ult} = 246 \text{ MPa}$, $t_{12} = 40 \text{ MPa}$).	10	L4	3	1
	b)	Explain the injection molding technique with a neat sketch. List its advantages and disadvantages.	10	L2	4	1
6.	a)	Discuss the Tsai-Wu Failure theory.	10	L2	3	1
	b)	Explain laser beam cutting of composites, with a neat sketch.	10	L2	4	1
		Unit – III				
7.	a)	Discuss the applications of polymer matrix composites in aircrafts and space systems.	10	L2	5	1
	b)	Explain the stir-casting process for manufacturing MMCs with a neat sketch.	10	L2	5	2
8.	a)	Explain the different types of metal matrix composites.	10	L2	5	2
	b)	Explain the future potential of composites.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
